Software Engineering – Lab 2

Agile Backlog Creation & Sprint Simulation in Jira

Project: Peer-to-Peer Carpooling & Expense Splitter(#26)

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1. Project Overview

The project is a peer-to-peer carpooling system that matches drivers and riders with overlapping routes and supports automated fuel-cost sharing. The Jira backlog was created from the functional requirements identified in Lab 1.

2. Epics

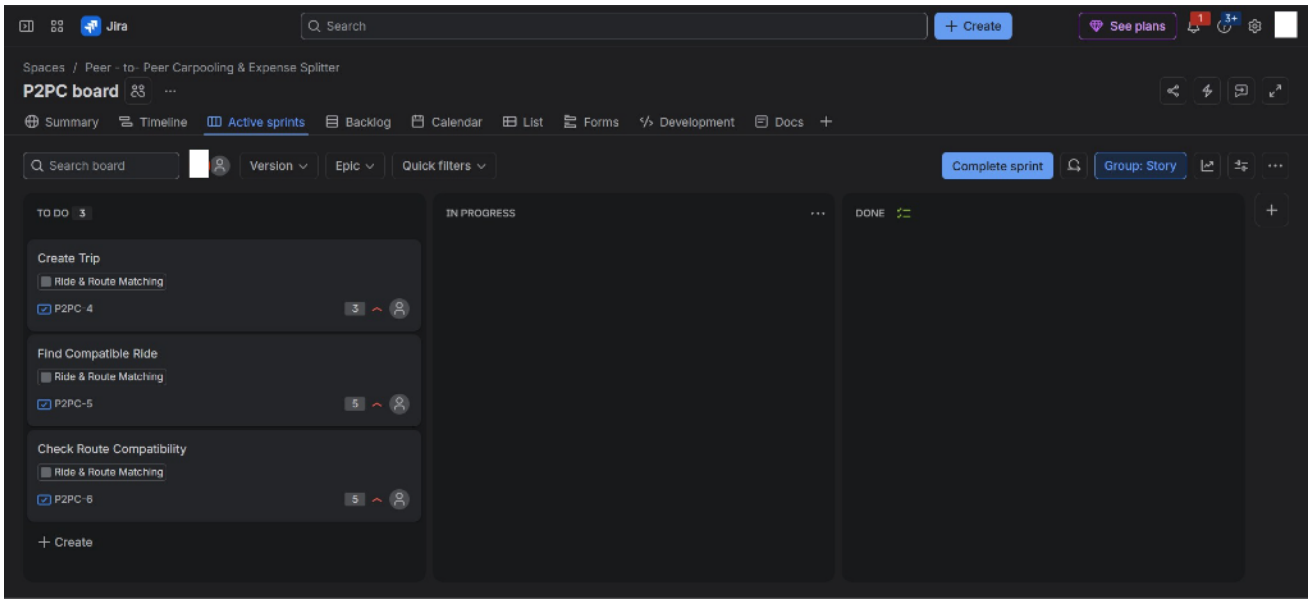
Epic	Purpose
Ride & Route Matching	Supports trip creation, finding compatible rides, and checking route compatib
Carpool Participation	Supports joining a suitable carpool and managing rider participation.
Expense Splitting	Supports calculating and splitting the fuel cost among participants.

3. User Stories and Estimation

ID	User Story	Priority	Points
P2PC-4	Create Trip	High	3
P2PC-5	Find Compatible Ride	High	5
P2PC-6	Check Route Compatibility	High	5
P2PC-7	Join Carpool	Medium	3
P2PC-8	Calculate Fuel Cost Split	High	5

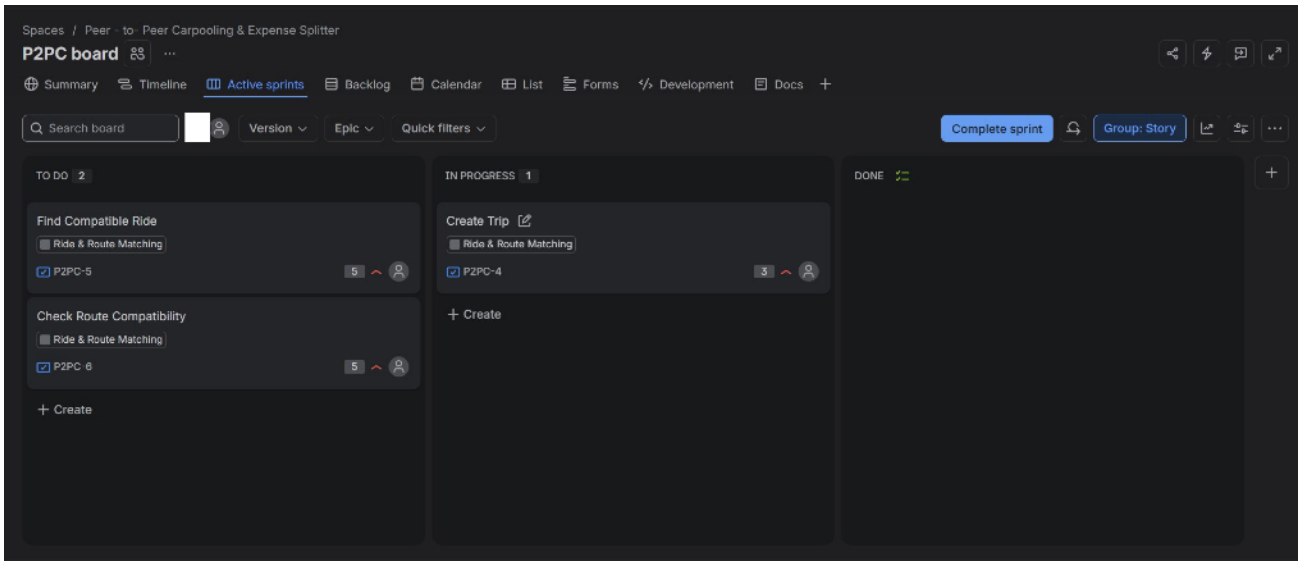
Story points were assigned using the Fibonacci-style scale. The total backlog estimate was 21 points: 13 points in Sprint 1 and 8 points in Sprint 2.

4. Sprint 1 – Initial Active Sprint Board



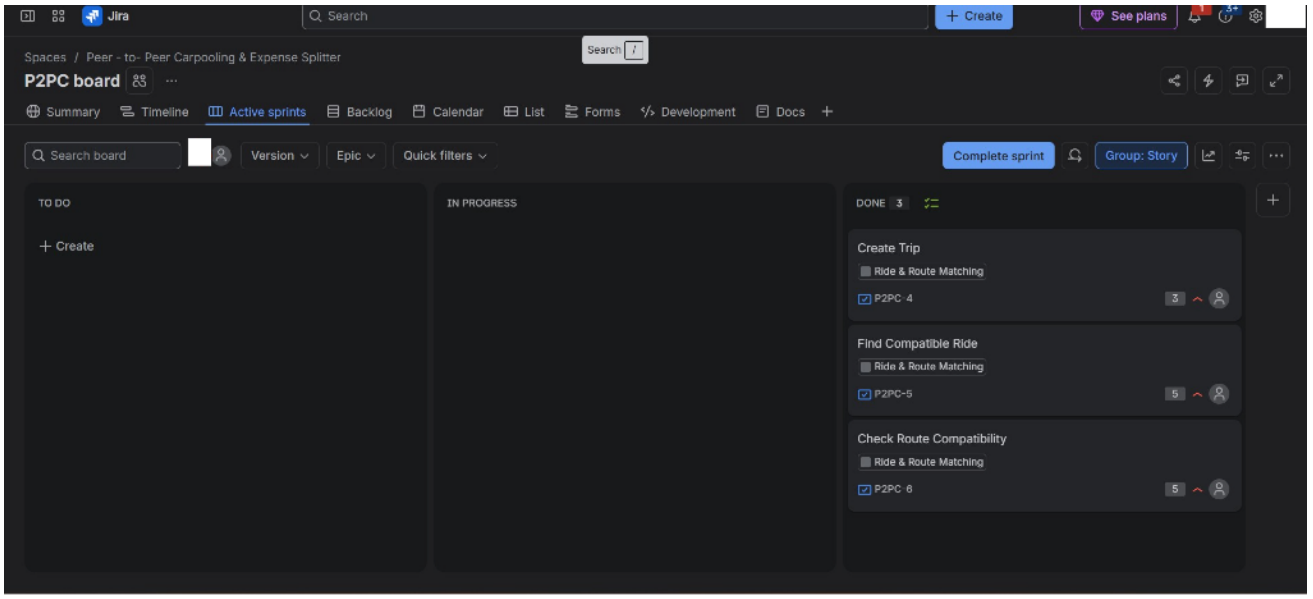
The first sprint was populated with the selected stories. The board shows the work items being tracked through the Scrum workflow.

5. Sprint 1 – Work Progress



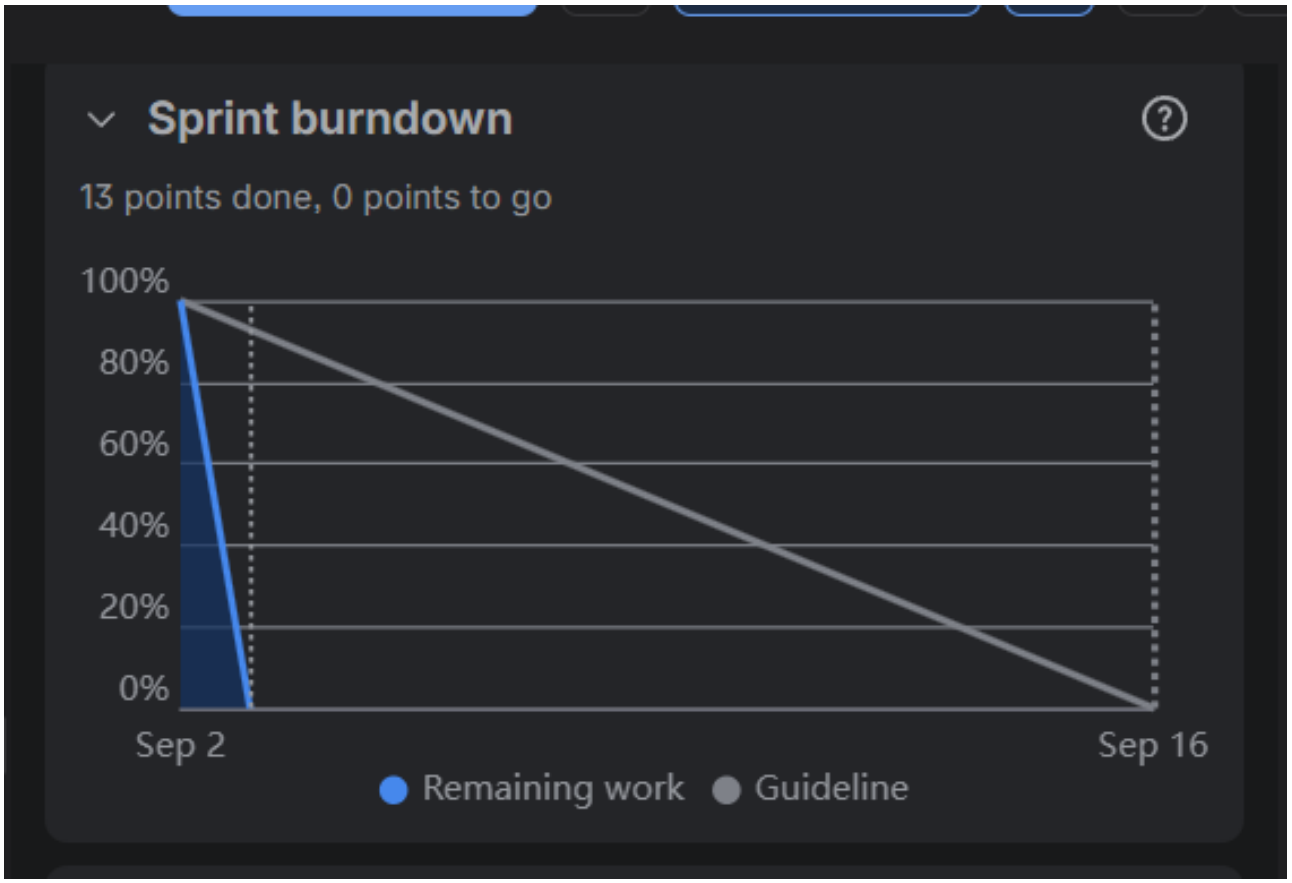
The stories were moved through the board as work progressed, demonstrating the To Do, In Progress and Done workflow.

6. Sprint 1 – Completed Work



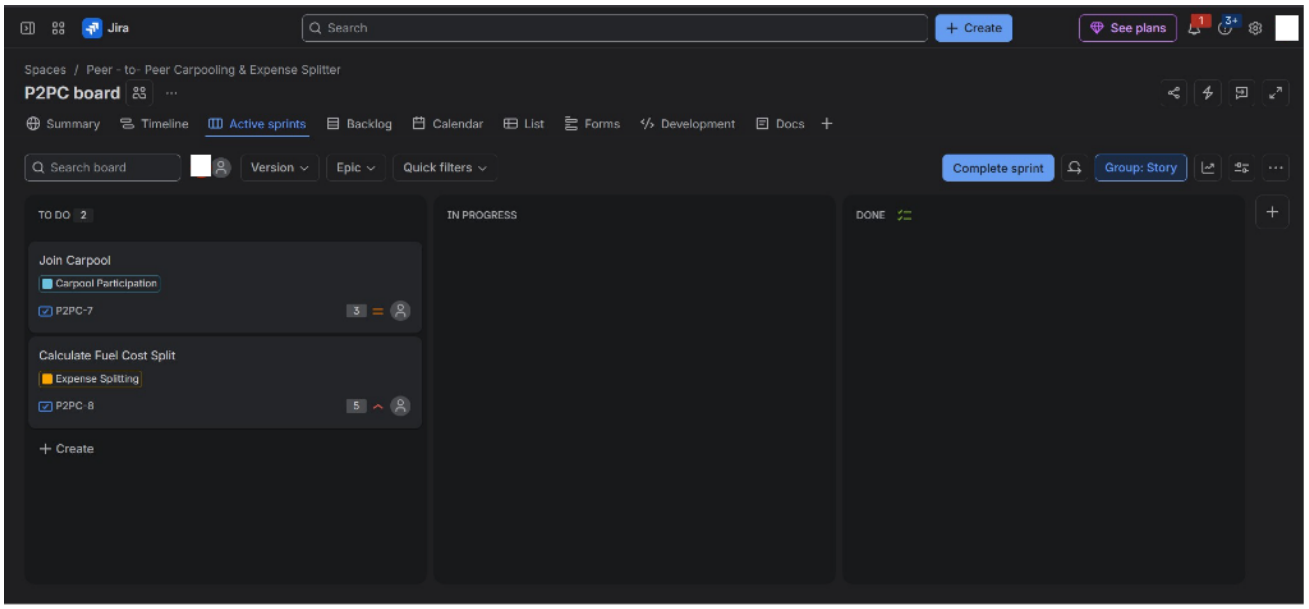
Sprint 1 was completed with the planned three stories finished. The completed work totalled 13 story points.

7. Sprint 1 – Burndown Chart



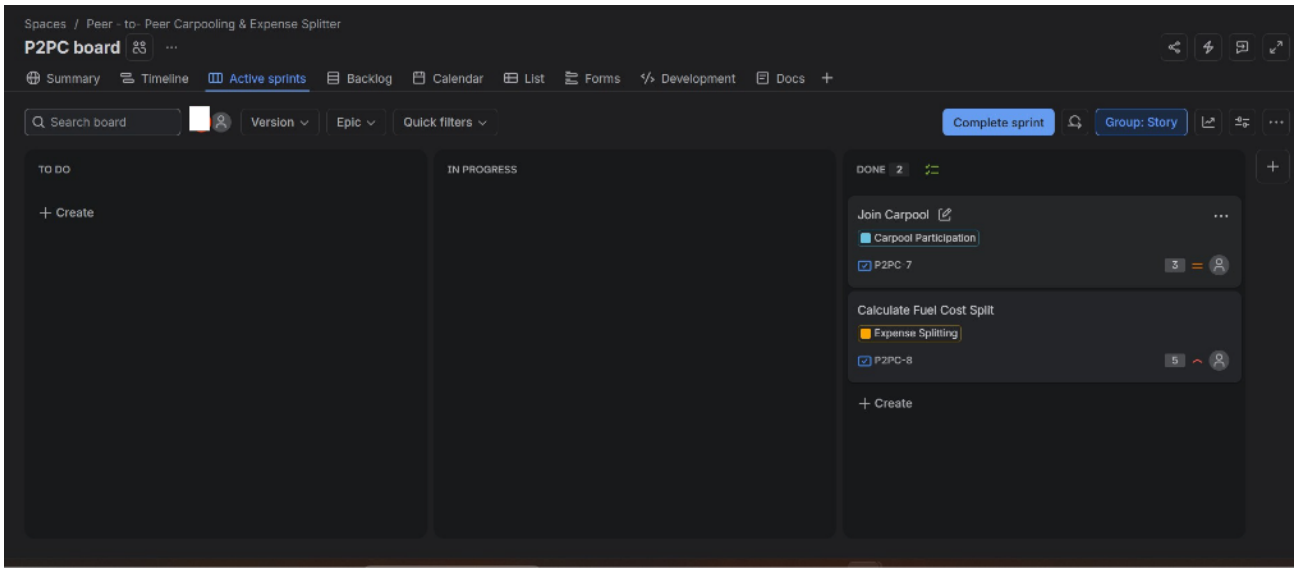
The burndown chart shows 13 points done and 0 points remaining. The remaining-work line drops to zero, indicating that the planned sprint work was completed.

8. Sprint 2 – Active Sprint Board



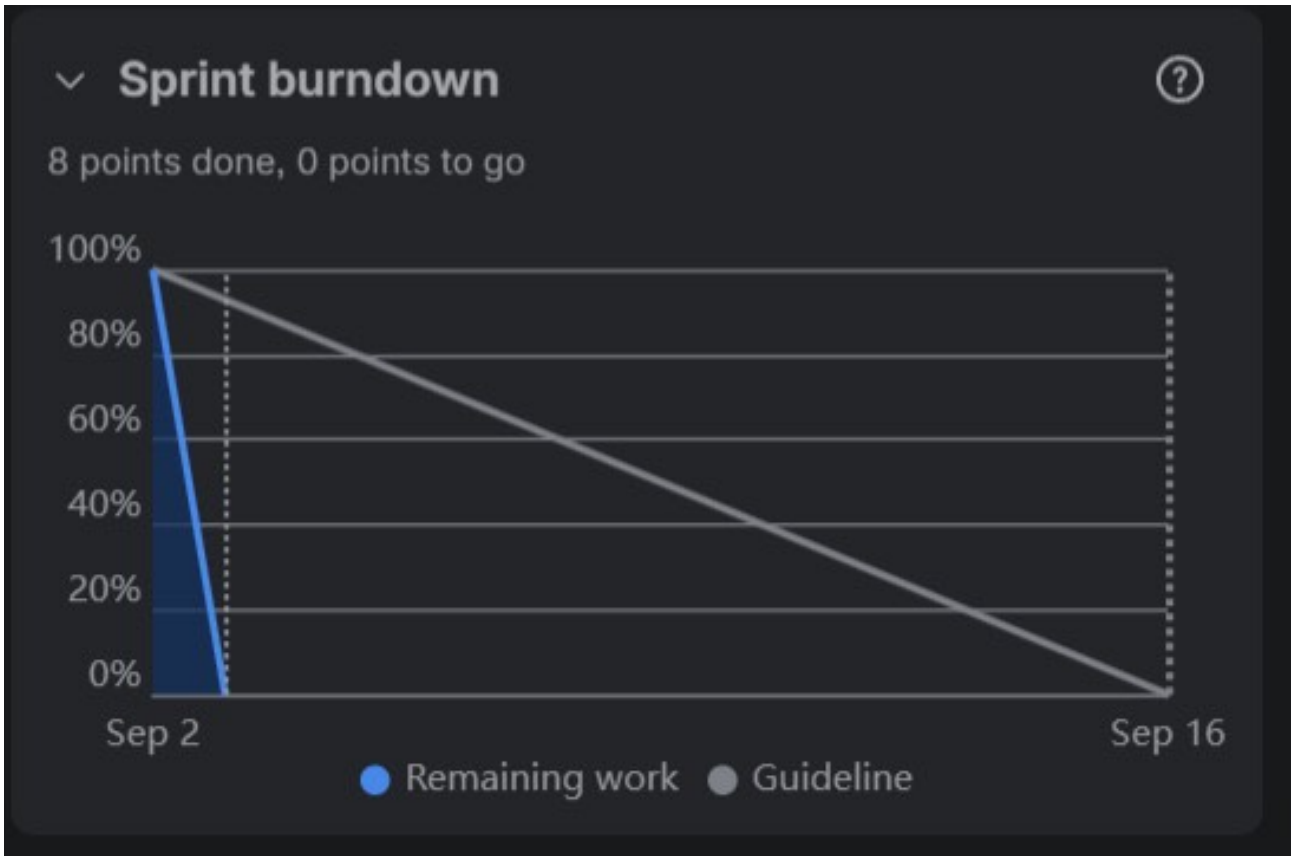
Sprint 2 contained the remaining two user stories: Join Carpool and Calculate Fuel Cost Split.

9. Sprint 2 – Completed Work



Both Sprint 2 stories were completed and moved to Done, giving a total of 8 completed story points.

10. Sprint 2 – Burndown Chart



The burndown chart shows 8 points done and 0 points remaining, confirming completion of the second sprint.

Sprint Reflection

Did your estimations match the actual effort? The estimates were reasonable for the mock sprint because all planned stories were finished with no pending work left over. Sprint 1 completed 13 points and Sprint 2 completed 8 points. Since this was a Jira simulation instead of real coding, the result mostly shows that the planned workload was easy to manage.

Was your backlog well-prioritised? Yes. The more important features—like creating trips, finding matching rides, and checking routes—were done first. Joining trips and splitting expenses were then finished in Sprint 2.

How did your simulated sprint match your plan? The sprints matched the plan because the chosen stories were completed inside their correct sprints. There was no spillover into another sprint.

What insights did the burndown chart give about your team's capacity? Both burndown charts hit zero remaining work. Sprint 1 finished 13 points and Sprint 2 finished 8 points. The sharp drop in remaining work is normal because the lab used a simulated sprint where tasks were moved across the board very quickly. This shows that the planned workload was achievable for the simulation.

Final Deliverables Summary

The Jira work finished for this lab includes the project epics, five prioritised user stories with story-point estimates, two completed Scrum sprints, active sprint board proof, and burndown charts for both sprints.

- **Sprint 1:** P2PC-4, P2PC-5, P2PC-6 — 13 points completed.
- **Sprint 2:** P2PC-7, P2PC-8 — 8 points completed.
- **Total:** 21 story points completed.